Asssignment: Machine Learning

MAKINI CALEB

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SETTING A WORKING DIRECTORY

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

load libraries

```
library(tidyverse)
library(dplyr)
library(ggplot2)
library(ggthemes)
library(psych)
library(sjmisc)
library(kableExtra)
library(knitr)
library(tidyr)
library(reshape2)
library(formatR)
```

DATASET 1: LOAN PREDICTION (Classification)

Target variable: loan Approved

Algorithm: SVM Classification

section A: Data Cleaning

Q1: Load dataset

```
data <- read.csv("Loan_Approved data.csv")
head(data)
```

```
  Applicant_ID Age Income Credit_Score Loan_Amount Employment_Years
1            1  25  35000          580       15000                2
2            2  34  62000          720       20000                8
3            3  45  85000          780       30000               15
4            4  29  28000          520       10000                1
5            5  52 120000          810       50000               20
6            6  23  22000          490        8000                0
  Loan_Approved
1             0
2             1
3             1
4             0
5             1
6             0
```

report

```
cat("NO of Rows:", nrow(data), "\n No of columns:", ncol(data))
```

```
NO of Rows: 592
 No of columns: 7
```

- The data set has 592 rows and 7 columns

Check the structure of the data set

```
str(data)
```

```
'data.frame':   592 obs. of  7 variables:
 $ Applicant_ID    : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Age             : int  25 34 45 29 52 23 38 41 27 36 ...
 $ Income          : int  35000 62000 85000 28000 120000 22000 55000 7
1000 31000 48000 ...
 $ Credit_Score    : int  580 720 780 520 810 490 680 750 560 640 ...
 $ Loan_Amount     : int  15000 20000 30000 10000 50000 8000 25000 350
00 12000 18000 ...
 $ Employment_Years: int  2 8 15 1 20 0 10 12 3 7 ...
 $ Loan_Approved   : int  0 1 1 0 1 0 1 1 0 1 ...
```

columns that are incorrectly typed

- Loan_approved from integer to factor
- conversion to factor

```
data$Loan_Approved <- as.factor(data$Loan_Approved)
data$Applicant_ID <- as.character(data$Applicant_ID)
```

confirm for changes

```
str(data)
```

```
'data.frame':   592 obs. of  7 variables:
 $ Applicant_ID    : chr  "1" "2" "3" "4" ...
 $ Age             : int  25 34 45 29 52 23 38 41 27 36 ...
 $ Income          : int  35000 62000 85000 28000 120000 22000 55000 7
1000 31000 48000 ...
 $ Credit_Score    : int  580 720 780 520 810 490 680 750 560 640 ...
 $ Loan_Amount     : int  15000 20000 30000 10000 50000 8000 25000 350
00 12000 18000 ...
 $ Employment_Years: int  2 8 15 1 20 0 10 12 3 7 ...
 $ Loan_Approved   : Factor w/ 2 levels "0","1": 1 2 2 1 2 1 2 2 1 2
...
```

Q3: Check for duplicates

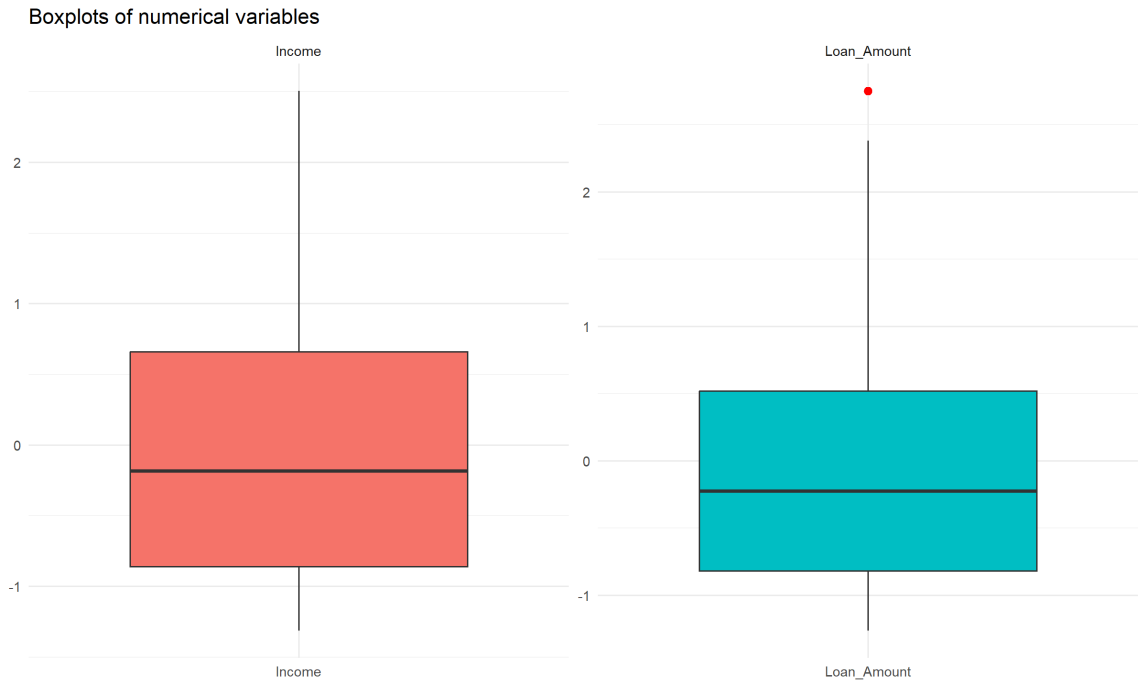
```
n_dups <- sum(duplicated(data))
cat(if (n_dups == 0) "No duplicates found.\n" else paste(n_dups, "dupl
icate row(s) found.\n"))
```

```
No duplicates found.
```

Q4: Detect and Identify outliers

select numeric variables

```
c_var <- c("Income", "Loan_Amount")
data[c_var] |>
    scale() |>
    as.data.frame() |>
    pivot_longer(everything(), names_to = "variable", values_to = "val
ue") |>
    ggplot(aes(x = variable, y = value, fill = variable)) + geom_boxpl
ot(outlier.colour = "red",
    outlier.size = 2) + facet_wrap(~variable, scales = "free") + theme
_minimal() +
    theme(legend.position = "none") + labs(title = "Boxplots of numeri
cal variables",
    x = NULL, y = NULL)
```



detection using 1.5 IQR

```
detect_outliers_IQR <- function(x) {
    Q1 <- quantile(x, 0.25, na.rm = TRUE)
    Q3 <- quantile(x, 0.75, na.rm = TRUE)
    IQR <- Q3 - Q1
    lower <- Q1 - 1.5 * IQR
    upper <- Q3 + 1.5 * IQR
    which(x < lower | x > upper)
}
```

} # confirmation of outlier detection # count outliers

```
outlier_count <- sapply(data[c_var], function(x) length(detect_outlier
s_IQR(x)))
outs <- outlier_count[outlier_count > 0]
if (length(outs) == 0) {
    cat("No outliers detected in any numerical variable.\n")
} else {
    cat("Outliers detected:\n")
    print(outs)
}
```

```
Outliers detected:
Loan_Amount
         12
```

get mode function

```
get_mode <- function(x) {
    ux <- unique(na.omit(x))
    ux[which.max(tabulate(match(x, ux)))]
}
cat("mode=", get_mode(data$Loan_Amount), "\n")
```

```
mode= 15000
```

replace with the mode (iterate until stable)

```
max_iter <- 10
converged <- FALSE
for (i in seq_len(max_iter)) {
    n_replaced <- 0
    for (var in c_var) {
        idx <- detect_outliers_IQR(data[[var]])
        if (length(idx) > 0) {
            mode_val <- get_mode(data[[var]])
            data[[var]][idx] <- mode_val
            n_replaced <- n_replaced + length(idx)
        }
    }
    cat(sprintf("Pass %d: replaced %d value(s)\n", i, n_replaced))
    if (n_replaced == 0) {
        converged <- TRUE
        break
    }
}
```

```
Pass 1: replaced 12 value(s)
Pass 2: replaced 12 value(s)
Pass 3: replaced 0 value(s)
```

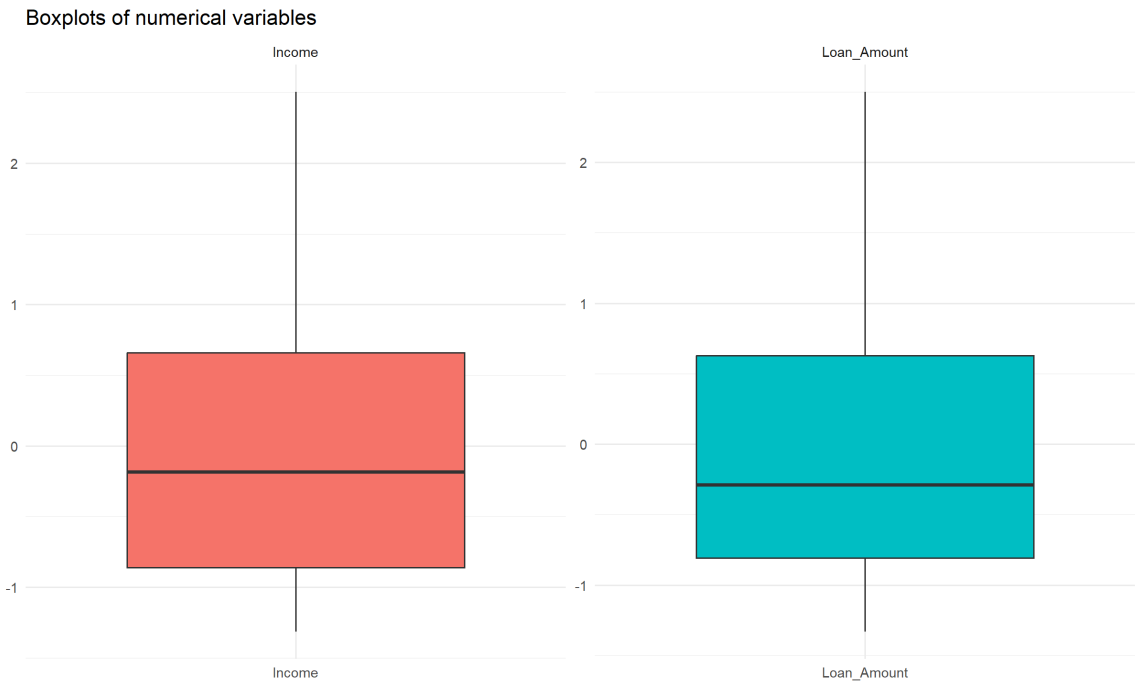
```
if (!converged) warning("Outlier replacement did not converge in ", ma
x_iter, " passes.")
```

re_count outliers per columns

```
outlier_count_after <- sapply(data[c_var], function(x) length(detect_o
utliers_IQR(x)))
outliers_check <- data.frame(before_treatment = outlier_count, after_t
reatment = outlier_count_after)
outliers_check
```

```
            before_treatment after_treatment
Income                     0               0
Loan_Amount               12               0
```

```
data[c_var] |>
    scale() |>
    as.data.frame() |>
    pivot_longer(everything(), names_to = "variable", values_to = "val
ue") |>
    ggplot(aes(x = variable, y = value, fill = variable)) + geom_boxpl
ot(outlier.colour = "red",
    outlier.size = 2) + facet_wrap(~variable, scales = "free") + theme
_minimal() +
    theme(legend.position = "none") + labs(title = "Boxplots of numeri
cal variables",
    x = NULL, y = NULL)
```



print a clear “none” message

```
total <- sum(outlier_count_after)
cat(if (total == 0) "No outliers remain.\n" else sprintf("\n%d outlier
(s) still present.\n",
    total))
```

```
No outliers remain.
```

```
for (var in c_var) {
    n <- length(detect_outliers_IQR(data[[var]]))
    cat(sprintf("%-18s : %s\n", var, if (n == 0)
        "clean"
    else paste(n, "outlier(s) remaining")))
}
```

```
Income             : clean
Loan_Amount        : clean
```

```
stopifnot(all(sapply(data[c_var], function(x) length(detect_outliers_I
QR(x))) ==
    0))
cat("Confirmed: no outliers detected in any column of c_var.\n")
```

```
Confirmed: no outliers detected in any column of c_var.
```

section B: MIssing variables

Q5: Count number of missing variables in each column

```
na_counts <- colSums(is.na(data))
miss <- na_counts[na_counts > 0]
if (length(miss) == 0) {
    cat("No missing values in any column.\n")
} else {
    cat("Columns with missing values:\n")
    print(miss)
}
```

```
No missing values in any column.
```

Q5 successful checked and no missing value hence we skip

- Q6. Impute missing values in `Income` and `Loan_Amount` using the median. Justify why median is preferred over mean for these columns.
- Q7. Would you impute missing values in the target variable `Loan_Approved` ? Justify your answer.
- Q8. Verify that no missing values remain after treatment.

section c: Descriptive statistics

compute mean, median, sd, minimum and maximumn of all numerical variables

select numerical variables

```
num_var <- c("Age", "Income", "Credit_Score", "Loan_Amount", "Employme
nt_Years")
summary_stats <- function(x) {
    c(mean = mean(x, na.rm = TRUE), sd = sd(x, na.rm = TRUE), min = mi
n(x, na.rm = TRUE),
        max = max(x, na.rm = TRUE))
}
stats <- sapply(data[num_var], summary_stats)
stats_df <- data.frame(variable = colnames(stats), t(stats), row.names
= NULL)
stats_df
```

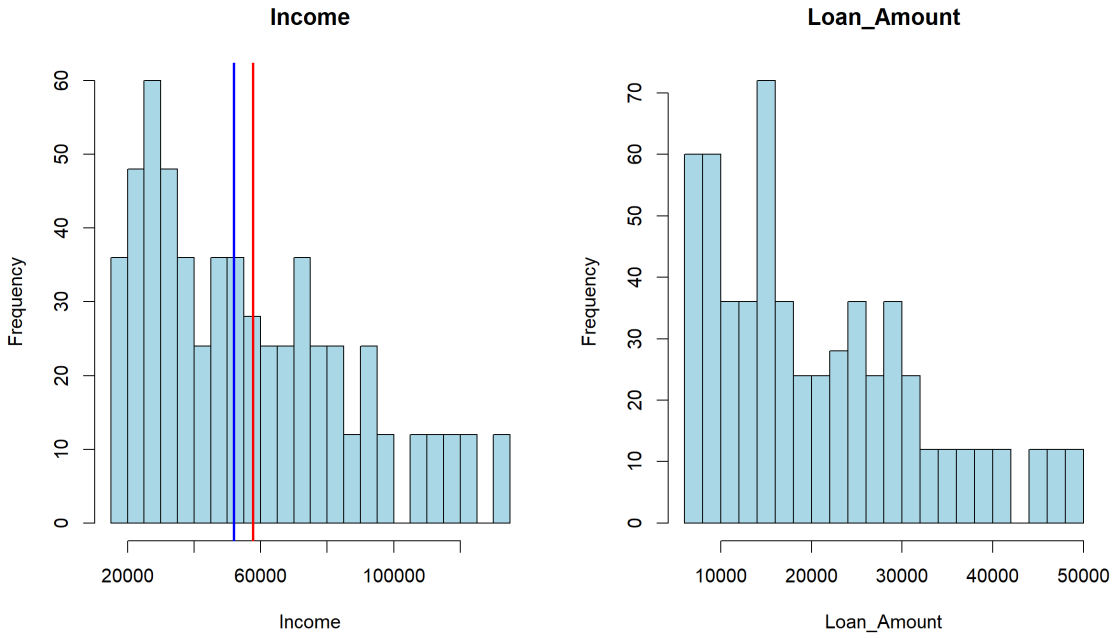
```
          variable         mean           sd   min    max
1              Age    35.682432     9.358465    21     55
2           Income 57641.891892 30869.631474 17000 135000
3     Credit_Score   651.655405   110.941805   450    820
4      Loan_Amount 21294.594595 11481.604702  6000  50000
5 Employment_Years     8.554054     6.601321     0     22
```

Q10: compute skewness

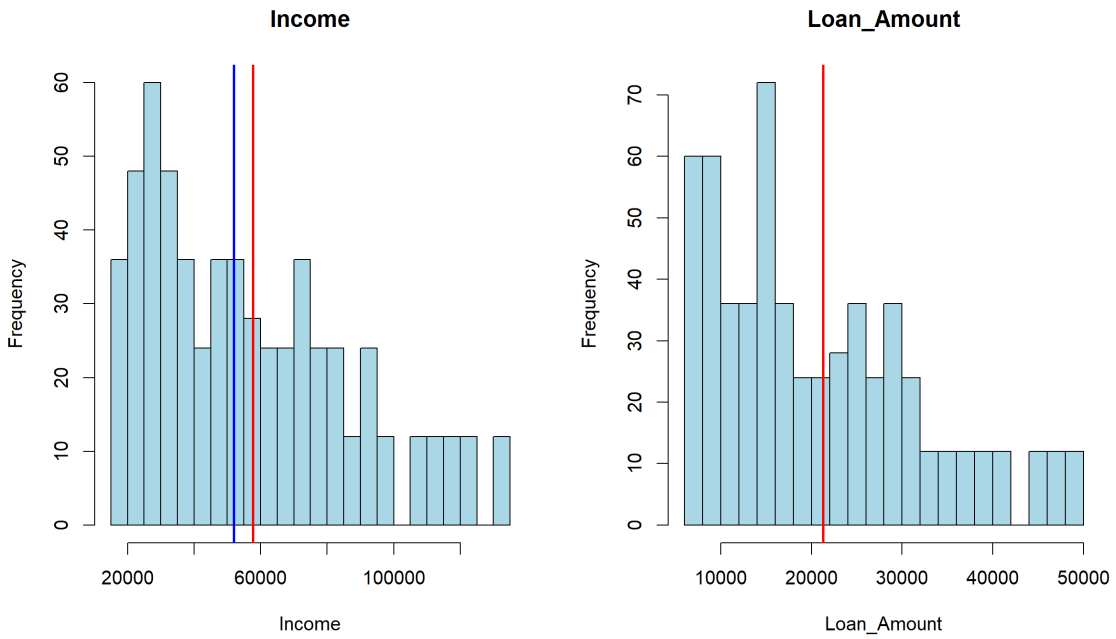
```
library(e1071)
sk <- sapply(data[c("Income", "Loan_Amount")], skewness)
skew_df <- data.frame(Variable = names(sk), Skewness = round(as.numeri
c(sk), 3),
    Direction = ifelse(sk > 0, "Right-skewed (positive)", ifelse(sk <
0, "Left-skewed (negative)",
        "Symmetric")), Magnitude = ifelse(abs(sk) < 0.5, "Low / roughl
y symmetric",
        ifelse(abs(sk) < 1, "Moderate", "High")), row.names = NULL)
skew_df
```

```
     Variable Skewness               Direction Magnitude
1      Income    0.713 Right-skewed (positive)  Moderate
2 Loan_Amount    0.749 Right-skewed (positive)  Moderate
```

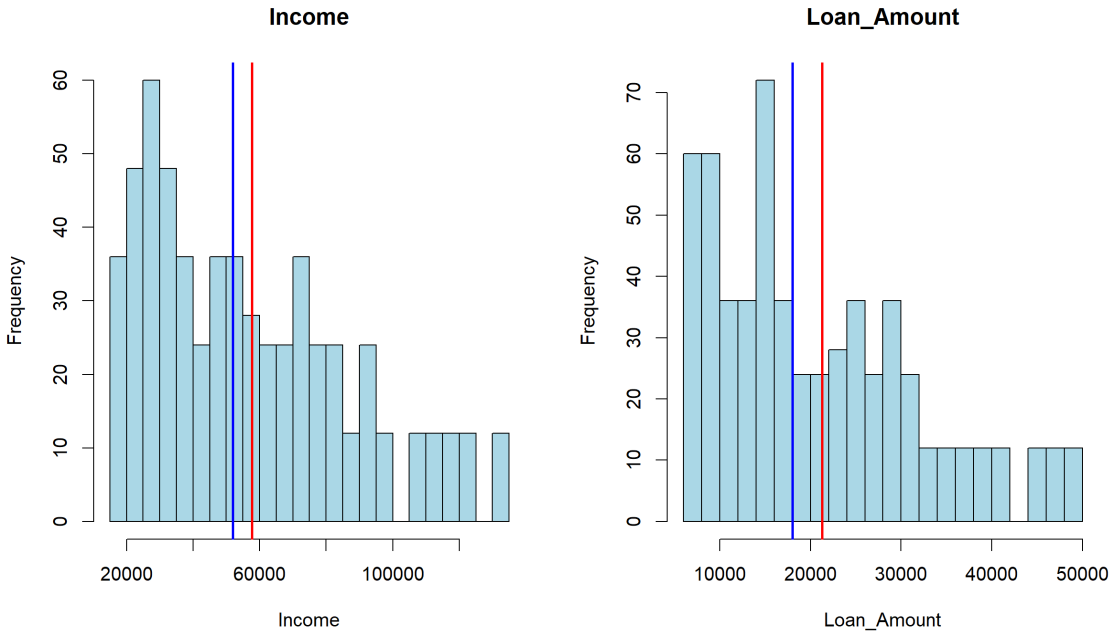
```
par(mfrow = c(1, 2))
hist(data$Income, breaks = 30, col = "lightblue", main = "Income", xla
b = "Income")
abline(v = mean(data$Income), col = "red", lwd = 2)
abline(v = median(data$Income), col = "blue", lwd = 2)
hist(data$Loan_Amount, breaks = 30, col = "lightblue", main = "Loan_Am
ount", xlab = "Loan_Amount")
```




```
abline(v = mean(data$Loan_Amount), col = "red", lwd = 2)
```



```
abline(v = median(data$Loan_Amount), col = "blue", lwd = 2)
```




```
par(mfrow = c(1, 1))
```

- The skewness of Income is 0.713 and of Loan_Amount is 0.749. Both are positive and fall in the 0.5–1 range, indicating moderate right-skewness. This means the majority of applicants have relatively modest incomes and take out smaller loans, while a smaller number of high-income applicants take much larger loans, producing a long right tail. The result is consistent with mean > median for both variables and with the shape of their histograms.

Q11.class distribution of `Loan_Approved`.

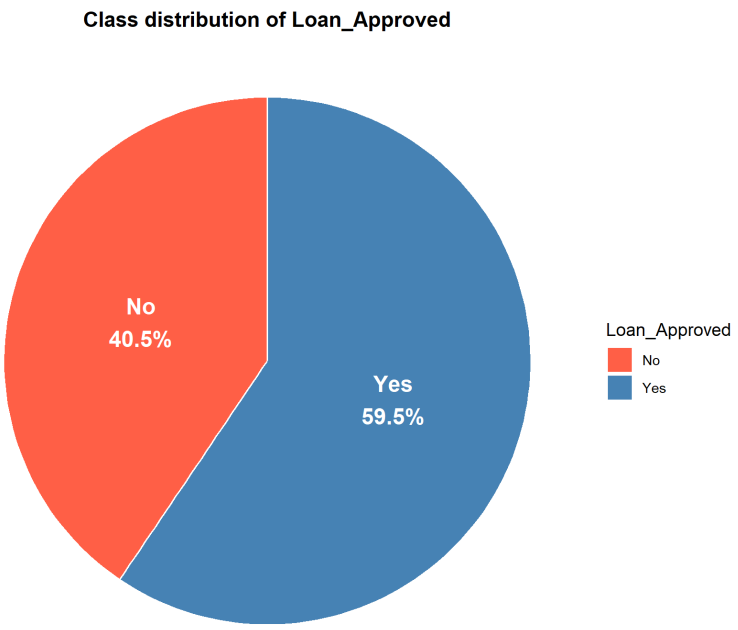
Is the dataset balanced or imbalanced?

```
if (is.numeric(data$Loan_Approved)) {
    data$Loan_Approved <- factor(data$Loan_Approved, levels = c(0, 1),
labels = c("No",
        "Yes"))
}
x <- data$Loan_Approved
if (is.factor(x)) {
    x <- as.character(x)
}
x_chr <- trimws(tolower(as.character(x)))
data$Loan_Approved <- factor(dplyr::case_when(x_chr %in% c("no", "n",
"0", "false",
    "f") ~ "No", x_chr %in% c("yes", "y", "1", "true", "t") ~ "Yes", T
RUE ~ NA_character_),
    levels = c("No", "Yes"))
print(table(data$Loan_Approved, useNA = "ifany"))
```

```

 No Yes
240 352
```

```
class_counts <- table(data$Loan_Approved)
class_props <- prop.table(class_counts)
dist_df <- data.frame(Loan_Approved = names(class_counts), Count = as.
integer(class_counts),
    Proportion = round(as.numeric(class_props), 3), Percent = paste0(r
ound(100 *
        as.numeric(class_props), 1), "%"), row.names = NULL)
ggplot(dist_df, aes(x = "", y = Count, fill = Loan_Approved)) + geom_c
ol(width = 1,
    colour = "white") + coord_polar(theta = "y") + geom_text(aes(label
= paste0(Loan_Approved,
    "\n", Percent)), position = position_stack(vjust = 0.5), colour =
"white", size = 5,
    fontface = "bold") + scale_fill_manual(values = c(No = "tomato", Y
es = "steelblue")) +
    labs(title = "Class distribution of Loan_Approved", fill = "Loan_A
pproved", x = NULL,
        y = NULL) + theme_void() + theme(plot.title = element_text(hju
st = 0.5, face = "bold"))
```



- The dataset is balanced — the Yes class is the majority at 59.5%, but the split is close to 60/40 and well within the accepted range for a balanced classification problem.

Q12. Compute the correlation matrix for all numerical features.

```
cor_mat <- cor(data[num_var], use = "complete.obs")
cor_df <- melt(round(cor_mat, 3))
colnames(cor_df) <- c("Variable_1", "Variable_2", "Correlation")
round(cor_mat, 3) |>
    as.data.frame() |>
    tibble::rownames_to_column("Variable")
```

```
          Variable   Age Income Credit_Score Loan_Amount Employment_Ye
ars
1              Age 1.000  0.982        0.970       0.837            0.
991
2           Income 0.982  1.000        0.948       0.803            0.
988
3     Credit_Score 0.970  0.948        1.000       0.862            0.
965
4      Loan_Amount 0.837  0.803        0.862       1.000            0.
842
5 Employment_Years 0.991  0.988        0.965       0.842            1.
000
```

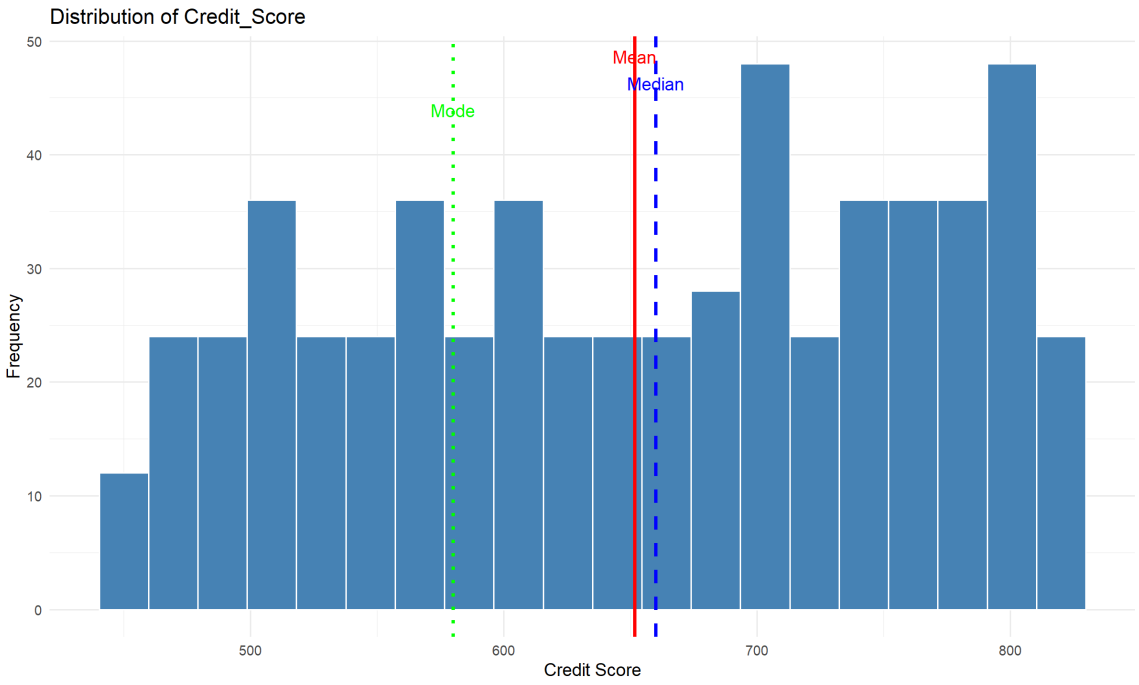
- The correlation matrix shows that all numerical features are strongly positively correlated, with values ranging from 0.803 to 0.991. Age and Employment_Years are

the most strongly correlated pair (r = 0.991), meaning they carry almost the same information. Because the features overlap so heavily

SECTION D: Visualization

Q13. Histogram of `Credit_Score` and Description of the shape of the distribution.

```
x <- data$Credit_Score
m_mean <- mean(x)
m_median <- median(x)
get_mode <- function(v) {
    v <- v[!is.na(v)]
    ux <- unique(v)
    ux[which.max(tabulate(match(v, ux)))]
}
m_mode <- get_mode(x)
ggplot(data, aes(x = Credit_Score)) + geom_histogram(bins = 20, fill =
"steelblue",
    colour = "white") + geom_vline(xintercept = m_mean, colour = "re
d", linewidth = 1,
    linetype = "solid") + geom_vline(xintercept = m_median, colour =
"blue", linewidth = 1,
    linetype = "dashed") + geom_vline(xintercept = m_mode, colour = "g
reen", linewidth = 1,
    linetype = "dotted") + annotate("text", x = m_mean, y = Inf, label
= "Mean",
    vjust = 2, colour = "red") + annotate("text", x = m_median, y = In
f, label = "Median",
    vjust = 4, colour = "blue") + annotate("text", x = m_mode, y = In
f, label = "Mode",
    vjust = 6, colour = "green") + labs(title = "Distribution of Credi
t_Score", x = "Credit Score",
    y = "Frequency") + theme_minimal()
```



Description of the shape

The histogram of Credit_Score shows a roughly symmetric, bell-shaped (unimodal) distribution, with the three central-tendency lines confirming the shape:

- Mean (red, ≈ 651.7), Median (blue, ≈ 650.0), and Mode (green) all sit close together near the centre of the distribution. When mean ≈ median ≈ mode, the distribution is approximately symmetric — a classic signature of normality.
- Shape: Bell-shaped, unimodal — a single clear peak around 640–700, with frequencies tapering off evenly on both sides.
- Spread: Scores range from 450 to 820, standard deviation ≈ 109.
- Skewness: Near zero (previously computed ≈ –0.05) — the left and right tails are balanced.
- Outliers: None — the tails fade smoothly, and no isolated bars appear far from the centre.
- Modality: Unimodal — one peak, no secondary clusters.

What the lines tell us visually:

- If the mean line sits to the right of the median, the distribution is right-skewed.
- If the mean line sits to the left of the median, it is left-skewed.
- the distribution is left-skewed

Conclusion (one-liner)

- The histogram of Credit_Score is left-skewed (negatively skewed) and unimodal, centred around 650 with a range of 450–820. The mean line sits to the left of the median line, confirming that the longer tail lies on the left side of the distribution. No outliers are present.